

MATH 13–Skeletal Notes

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Preface

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the “PDF” button in the navigation bar gives you the whole book as one file to print from. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

Everything for this course lives here: the notes themselves, the assignments, the review problems for each exam, and the solutions. Each item is added as we reach it, so check back regularly rather than assuming a page is final. Solutions to an assignment are posted only after its due date has passed; the review sets are posted the same way.

The problems are all gathered in the Exercises chapter. There are ten assignments, which are graded, and four sets of review problems, which are not — the review sets collect everything the assignments left over, and they are fair game on an exam. Assignments are due at 11:59 PM on the day shown below.

Assignments	Due day, at 11:59 PM
1, 2, 4, 5, 7, 8, 10	Friday
3, 6, 9	Wednesday (exam weeks 3, 6, and 9)

You hand your work in through Gradescope, not through this site. The steps — how to upload a scan or a photo of your written solutions, and how to tell Gradescope which page holds which problem — are under [Submitting Your Assignments in Gradescope](#) in the course syllabus. Read them once before the first assignment is due; a submission with the pages mismatched is the most common way points get lost for reasons that have nothing to do with the mathematics.

One more thing worth knowing before you start: each of the ten assignment pages carries a Socratic AI tutor that will coach you when you are stuck, without ever giving an answer away. How it works, what it will and will not do, and what to keep out of the chat are all set out under Using the AI Tutor, at the top of the Exercises chapter.

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Chapter 1

Hyperbolic Functions

Objectives

After completing this section, you should be able to:

- Define the hyperbolic sine and cosine in terms of e^x and e^{-x} , and differentiate them.
- Sketch the graphs of $\sinh x$ and $\cosh x$, and state their domains and ranges.
- Use the identity $\cosh^2 x - \sinh^2 x = 1$ and relate it to the parametrization of the hyperbola $x^2 - y^2 = 1$.
- Work with the remaining hyperbolic functions $\tanh x$, $\coth x$, $\operatorname{sech} x$, and $\operatorname{csch} x$, and apply hyperbolic identities to integrals and equations.

This chapter is about certain combinations of e^x and e^{-x} which appear in multiple branches of science and engineering. In many ways they are analogous to the trigonometric functions, and they have a similar relationship to the hyperbola that the trigonometric functions have to the circle, which is why they are known as **hyperbolic functions**. We will study these relationships gradually during this chapter.

These are skeletal notes: the definitions, identities, and problem statements are given, and blank boxes and empty grids are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1 Definitions and Derivatives

The **hyperbolic cosine** is defined as

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad (1.1.1)$$

and the **hyperbolic sine** is defined as

$$\sinh x = \frac{e^x - e^{-x}}{2}. \quad (1.1.2)$$

Find the derivatives $\frac{d}{dx}(\cosh x)$ and $\frac{d}{dx}(\sinh x)$ by differentiating the two definitions.

workspace

1.2 Graph, Domain, and Range of $\cosh x$ and $\sinh x$

Since $\sinh x$ and $\cosh x$ are built from the two exponentials $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$, their graphs are easiest to understand by sketching those exponentials first. See [Figure 1.2.1](#) and [Figure 1.2.2](#).

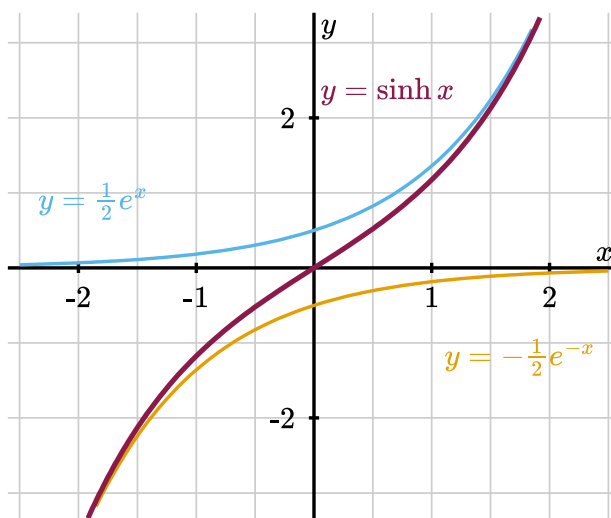


Figure 1.2.1 The graph of $y = \sinh x$. The curve is squeezed between the two exponentials $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$, approaching the first as $x \rightarrow \infty$ and the second as $x \rightarrow -\infty$.

From the graph, read off the domain, the range, and the symmetry of $\sinh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\sinh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $-\frac{1}{2}e^{-x}$ when x is large and negative. Confirm this observation.

workspace

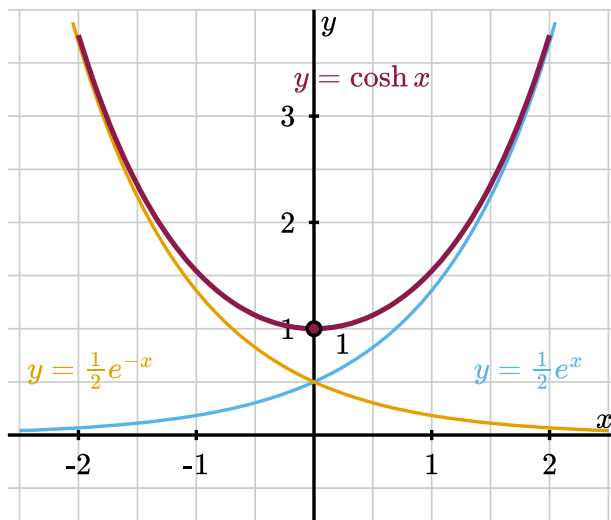


Figure 1.2.2 The graph of $y = \cosh x$. The curve is the sum of the two exponentials $y = \frac{1}{2}e^x$ and $y = \frac{1}{2}e^{-x}$, so it lies above both and has its minimum value 1 at $x = 0$.

From the graph, read off the domain, the range, and the symmetry of $\cosh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\cosh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $\frac{1}{2}e^{-x}$ when x is large and negative, while never dropping below 1. Confirm these observations.

workspace

1.3 Identities and Other Hyperbolic Functions

A similar identity to the trigonometric identity $\cos^2 x + \sin^2 x = 1$ holds for the hyperbolic functions:

$$\cosh^2 x - \sinh^2 x = 1. \quad (1.3.1)$$

As you already know, any point $P(x, y)$ on the circumference of the unit circle ($x^2 + y^2 = 1$) can be described in terms of sine and cosine of an angle $t \in [0, 2\pi)$, i.e. $x = \cos t$ and $y = \sin t$, which results in the trigonometric identity $\cos^2 t + \sin^2 t = 1$. Mark this on the circle in [Figure 1.3.1](#).

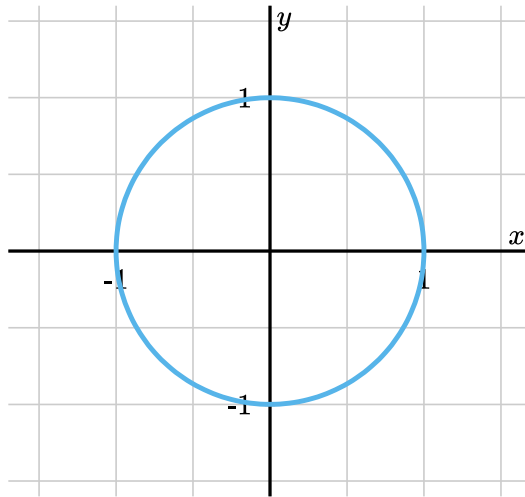


Figure 1.3.1 The unit circle $x^2 + y^2 = 1$. Mark a point $P(\cos t, \sin t)$ on it, and draw the right triangle with legs x and y and hypotenuse 1.

Similarly, any point $P(x, y)$ on the right branch of the hyperbola $x^2 - y^2 = 1$ can be represented as $x = \cosh t$ and $y = \sinh t$, where $t \in \mathbb{R}$. This follows directly from identity (1.3.1), i.e. $\cosh^2 t - \sinh^2 t = 1$. Mark this on the hyperbola in [Figure 1.3.2](#).

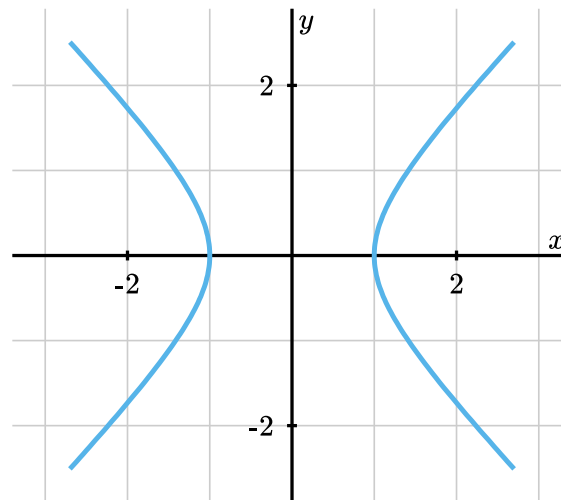


Figure 1.3.2 The hyperbola $x^2 - y^2 = 1$. Mark a point $P(\cosh t, \sinh t)$ on the right branch, and draw the segments from the origin to P and to the vertex.

Checkpoint 1.3.3 Parametrizing the Left Branch. Both branches of $x^2 - y^2 = 1$ appear in Figure 1.3.2, but the parametrization $P(\cosh t, \sinh t)$ traces only the right one, since $\cosh t \geq 1$ for every $t \in \mathbb{R}$. How would you parametrize the *left* branch, where $x \leq -1$?

Your answer and reasoning.

workspace

Hint. The identity $\cosh^2 t - \sinh^2 t = 1$ is unaffected if you change the sign of the first coordinate.

Checkpoint 1.3.4 True or False. The parametrization $x = -\cosh t$, $y = -\sinh t$, $t \in \mathbb{R}$, also traces the left branch of $x^2 - y^2 = 1$.

True or false? Your answer and reasoning.

workspace

As you might have guessed, the rest of the hyperbolic functions are defined

as follows.

$$\begin{aligned}\tanh x &= \frac{\sinh x}{\cosh x} & \operatorname{sech} x &= \frac{1}{\cosh x} \\ \coth x &= \frac{\cosh x}{\sinh x} & \operatorname{csch} x &= \frac{1}{\sinh x}\end{aligned}$$

1.4 Hyperbolic Identities

Below are some identities that you may find useful in some problems, however, you are not expected to memorize them. We will prove some of them as an exercise later. Each hyperbolic identity is listed next to the trigonometric identity it resembles.

Table 1.4.1 Hyperbolic identities and their trigonometric counterparts.

Hyperbolic	Trigonometric
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$
$\sinh 2x = 2 \sinh x \cosh x$	$\sin 2x = 2 \sin x \cos x$
$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\cosh^2 x = \frac{\cosh 2x + 1}{2}$	$\cos^2 x = \frac{1 + \cos 2x}{2}$
$\sinh^2 x = \frac{\cosh 2x - 1}{2}$	$\sin^2 x = \frac{1 - \cos 2x}{2}$

Notice the pattern: each hyperbolic identity is its trigonometric counterpart with the sign changed wherever two sines (or two tangents) are multiplied together.

Example 1.4.2 Computing $\cosh^{-1}(x)$. Show that the inverse hyperbolic cosine can be written in terms of the natural logarithm as

$$\cosh^{-1}(x) = \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1.$$

Solution. (Restrict $\cosh$ to $x \geq 0$ so that it is one-to-one, set $y = \cosh(x)$, and solve for x .)

workspace



Example 1.4.3 An integral via a hyperbolic substitution. Use hyperbolic functions to calculate the integral

$$\int \frac{\sqrt{x^2 - 1}}{x^2} dx, \quad x \geq 1.$$

Hint: Similar to $\int \sec^2(x) dx = \tan(x) + C$, we have $\int \operatorname{sech}^2(x) dx = \tanh(x) + C$.

Solution.

workspace



1.5 Hyperbolic Functions in Real Life

Hyperbolic functions are more than a formal analogue of the trigonometric functions: they describe a number of concrete physical situations. We look at three of them.

1.5.1 *Hanging Cables and the Catenary

Starred section. This one is for the interested reader. It will not be examined.

If a heavy flexible cable (such as a telephone line, a power line, or a chain) hangs freely from two supports, it settles into a curve called a **catenary**. Contrary to a common guess, this curve is not a parabola; it is the graph of a hyperbolic cosine,

$$y = a \cosh\left(\frac{x}{a}\right),$$

where x is measured horizontally from the lowest point of the cable, so that the y -axis is the axis of symmetry and the lowest point sits at height $y(0) = a$. Sliding the curve up or down, as in $y = c + a \cosh(x/a)$, only changes where we draw the x -axis.

The shape comes out of a balance of forces. Look at the piece of cable running from the lowest point to a point where the cable makes an angle θ with the horizontal, and let s be the arc length of that piece. Three forces act on it: the tension T_0 at the lowest point, which is horizontal; the tension T

along the cable at the other end; and the weight $\lambda g s$ of the piece, where λ is the mass per unit length; see Figure 1.5.1. Balancing the horizontal and the vertical components gives

$$\begin{aligned} T \cos \theta &= T_0 \\ T \sin \theta &= \lambda g s. \end{aligned}$$

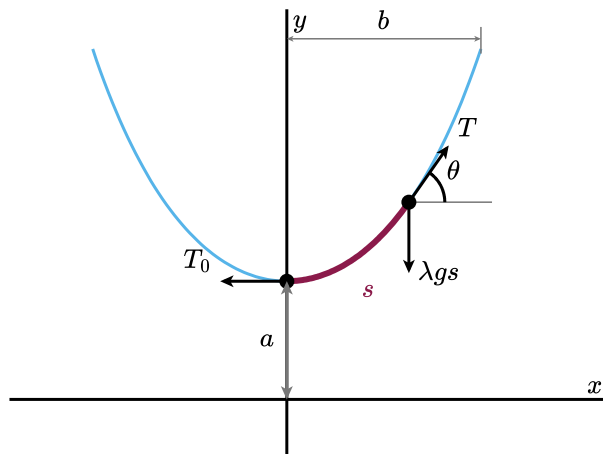


Figure 1.5.1 The three forces on the piece of cable of arc length s running from the lowest point of the cable to a point where the cable makes an angle θ with the horizontal. The lowest point sits at height $a = T_0/(\lambda g)$, and the supports are a distance $2b$ apart. (After Fig. 1 of Behroozi, cited below.)

The quantity $a = T_0/(\lambda g)$ has units of length, and dividing the second equation by the first eliminates T and leaves $s = a \tan \theta$. Since the cable makes the angle θ with the horizontal, $\tan \theta$ is its slope, so

$$\frac{dy}{dx} = \tan \theta = \frac{s}{a}.$$

Now use the arc length element $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (s/a)^2} dx$ and separate the variables:

$$dx = \frac{a ds}{\sqrt{a^2 + s^2}} \quad \Longrightarrow \quad x = a \sinh^{-1} \left(\frac{s}{a} \right),$$

where the constant of integration vanishes because $s = 0$ at $x = 0$.

Solving for s gives $s = a \sinh(x/a)$, and therefore

$$\frac{dy}{dx} = \frac{s}{a} = \sinh \left(\frac{x}{a} \right).$$

One last integration, together with $y(0) = a$, produces the catenary equation $y = a \cosh(x/a)$.

Two things are worth noticing. First, a is a pure scale factor: written as $y/a = \cosh(x/a)$, the equation shows that every catenary is a scaled copy of the single curve $y = \cosh x$, in exactly the same way that every circle is a scaled copy of the unit circle; Figure 1.5.2 shows four of them. Second, a is fixed by the cable itself. If the cable has half-length l and its two supports are a distance $2b$ apart, then putting $s = l$ at $x = b$ in $s = a \sinh(x/a)$ gives

$$l = a \sinh \left(\frac{b}{a} \right),$$

which determines a (numerically) from the two lengths. Since $a = T_0/(\lambda g)$, a cable pulled tight has a large a and hangs almost flat, while a slack one has a small a and sags sharply.

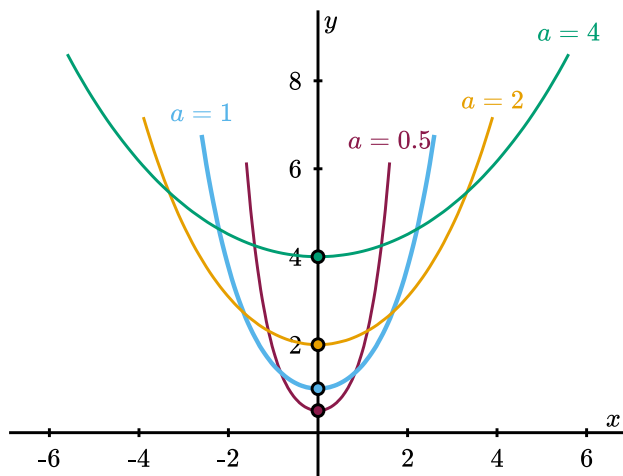


Figure 1.5.2 The catenaries $y = a \cosh(x/a)$ for $a = 0.5, 1, 2, 4$. Each curve meets the y -axis at its own value of a , and all four are scaled copies of the single curve $y = \cosh x$. (After Fig. 2 of Behroozi, cited below.)

A worked example of a hanging cable, in which we find the slope of the cable and the angle at which it meets its pole, appears in [Example 1.6.1](#).

The derivation above follows F. Behroozi, [In Praise of the Catenary](#), *The Physics Teacher* **56**, 214–217 (2018), which also discusses the sense in which all catenaries are similar to one another and suggests simple classroom demonstrations.

1.5.2 Celestial Mechanics

If a comet has enough speed, it can escape the gravitational pull of the sun, in which case one possible trajectory is a hyperbolic trajectory. The comet [2I/Borisov](#), discovered in 2019, is the first comet known to have come from outside our solar system. It was moving too fast for the sun to capture it, so its path is a hyperbola rather than an ellipse: it swung around the sun once and is now on its way back out. See [Figure 1.5.3](#).

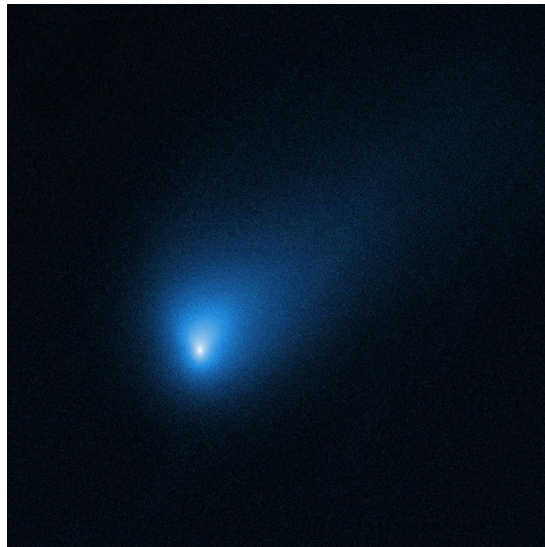


Figure 1.5.3 The interstellar comet 2I/Borisov, photographed by the Hubble Space Telescope in 2019. Because its speed exceeds the escape speed of the sun, its trajectory is a hyperbola and it passes through the solar system only once. (Image: NASA, ESA and D. Jewitt (UCLA).)

1.5.3 The Gateway Arch

The Gateway Arch in St. Louis, Missouri (designed in 1963 and completed in 1965) is a catenary turned upside down: flipping the curve converts the tension carried by a hanging chain into pure compression, which is what masonry and steel carry best. The geometric form of the gateway was set by Hannskari Bandel (structural engineer) and was expressed in the blueprints by the equation

$$y = A \left(\cosh \frac{Cx}{L} - 1 \right),$$

where A , C , and L are constants. The arch is slightly flattened compared with a uniform hanging chain, because it is thicker at the base than at the top. It stands 630 feet tall and 630 feet wide at the base; the National Park Service describes its construction and its geometry at [Gateway Arch National Park](#). See [Figure 1.5.4](#).



Figure 1.5.4 The Gateway Arch in St. Louis, Missouri. Its centerline follows the curve $y = A(\cosh(Cx/L) - 1)$, an upside-down catenary. (Photograph by John Margolies, 1988; John Margolies Roadside America photograph archive, Library of Congress, Prints and Photographs Division.)

1.6 Further Examples

Example 1.6.1 A hanging telephone line. A telephone line hangs between two poles 14 m apart in the shape of the catenary $y = 20 \cosh(x/20) - 15$, where x and y are measured in meters.

- A. Find the slope of this curve where it meets the right pole.
- B. Find the angle θ between the line and the pole.

Solution. (A sketch of the line between the poles at $x = -7$ and $x = 7$ may help.)

workspace

Example 1.6.2 Rewriting $\sinh(\ln(x))$. Consider the function $f(x) = \sinh(\ln(x))$. ▲

A. Express $f(x)$ as a fraction of two polynomials.

B. Calculate $\frac{df}{dx}$.

Solution.

workspace

▲

Example 1.6.3 A double-angle identity. Prove that $\sinh(2x) = 2 \sinh(x) \cosh(x)$.

Proof (start from the definitions).

workspace



Example 1.6.4 Solving a hyperbolic equation. Consider the equation $2 \cosh(2x) + 10 \sinh(2x) = 5$ and solve for x .

Solution.

workspace



Chapter 2

Series and Sequences

2.1 Infinite Series

Objectives

After completing this section, you should be able to:

- Define an infinite sequence, an infinite series, and the sequence of partial sums, and decide convergence of a series from its partial sums.
- Recognize a geometric series, and determine whether it converges and what it converges to.
- Apply the n th Term Test for divergence, and understand what the test can and cannot conclude.
- Combine convergent series using the sum, difference, and constant multiple rules, and evaluate telescoping series from their partial sums.

What does it mean to add infinitely many numbers together? In this section we answer that question by defining an infinite series and its sequence of partial sums, so that convergence of a series becomes a statement about a limit we already know how to study. We then put the definition to work on the geometric series, whose behavior we can settle completely, and on the n th Term Test, which is our first quick check for divergence.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

2.1.1 Definitions

Definition 2.1.1 Sequence. A **sequence** is a list of numbers, $a_1, a_2, a_3, \dots$. An **infinite sequence** of numbers is a function whose domain is the set of positive integers. ◆

Example 2.1.2 A Sequence of Halves. The numbers $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$ form an infinite sequence; its n th term is $a_n = \left(\frac{1}{2}\right)^n$. ▲

Definition 2.1.3 Infinite Series. The sum of the numbers in an infinite

sequence $\{a_n\}$, i.e. $a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$, is called an **infinite series**.

Here a_n is the n th term of the series. ◆

Example 2.1.4 An Infinite Series with a Finite Sum. Infinite sequences can have finite sums. Consider the sum of the sequence from [Example 2.1.2](#), i.e. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$.

It is most convenient to evaluate the result of this sum geometrically. Draw a square of side one in the space below. Shade half of it, then half of what is left, then half of what is left after that, and keep going. Label the pieces $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$. What is the total shaded area, and therefore what is the sum?

draw the unit square and its subdivisions here



2.1.2 Partial Sums

Consider again the infinite sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$. Let us denote the sum of the first n terms in this sequence by s_n , which are known as **partial sums**. Can we find a pattern in the sequence of partial sums?

Write out s_1 , s_2 , and s_3 , each one both as a fraction and in the form $1 - \square$.

Then guess a closed formula for $s_n = \sum_{k=1}^n \left(\frac{1}{2}\right)^k$.

$$s_1 =$$

$$s_2 =$$

$$s_3 =$$

$$\vdots$$

$$s_n =$$

Now use your formula for s_n to compute the infinite series, by taking the limit of the partial sum as $n \rightarrow \infty$. Compare the answer with the one you found geometrically from the square.

workspace

In the example above, the sequence of partial sums converged to a number. In general, if the sequence of the partial sums converges to a number, we say that the series **converges**, otherwise we say that the series **diverges**.

2.1.3 Geometric Series

An important example of infinite series is the geometric series. The **geometric series** is of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}, \quad (2.1.1)$$

where a, r are real numbers and $a \neq 0$. Note that we can re-write the series as

$$\sum_{n=0}^{\infty} ar^n.$$

Let us compute the partial sum s_n for the geometric series, so that

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}. \quad (2.1.2)$$

Multiply (2.1.2) by r , subtract the result from (2.1.2), and solve for s_n . Almost every term should cancel.

$$r s_n =$$

$$s_n - r s_n =$$

$$s_n =$$

Next, take the limit of your partial sum as $n \rightarrow \infty$. Treat the cases $|r| < 1$, $|r| > 1$, and $r = 1$ separately, and in each case say what $\lim_{n \rightarrow \infty} r^n$ does.

$$|r| < 1:$$

$$|r| > 1:$$

$$r = 1:$$

Checkpoint 2.1.5 What Happens When $r = -1$? We have now handled $|r| < 1$, $|r| > 1$, and $r = 1$, but one case is still missing: $r = -1$. Write out the partial sums s_1, s_2, s_3, s_4 of the series $a - a + a - a + \cdots = \sum_{n=1}^{\infty} a(-1)^{n-1}$ for $a \neq 0$. Does the sequence of partial sums approach a single number as $n \rightarrow \infty$? What does that tell you about the series?

Your answer and reasoning.

workspace

Hint. The partial sums do not grow without bound here, the way they do

when $r = 1$. Look instead at whether they settle down to one value.

Putting all of the cases together, we can say that if $|r| \geq 1$, the geometric series is divergent.

Summary.

The geometric series (2.1.1) converges to $\frac{a}{1-r}$ if $|r| < 1$, i.e.

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1, \quad (2.1.3)$$

and diverges if $|r| \geq 1$.

Next, as an application of the geometric series, we will go through the following example, which is from our textbook.

Example 2.1.6 A bouncing ball. You drop a ball from a meters above a flat surface. Each time the ball hits the surface after falling a distance h , it rebounds a distance rh , where r is positive but less than 1. Find the total distance the ball travels up and down.

Solution. (Use Figure 2.1.7 to write the total distance as an infinite series, then match it against (2.1.3). Watch the first drop: it is travelled only once.)

workspace



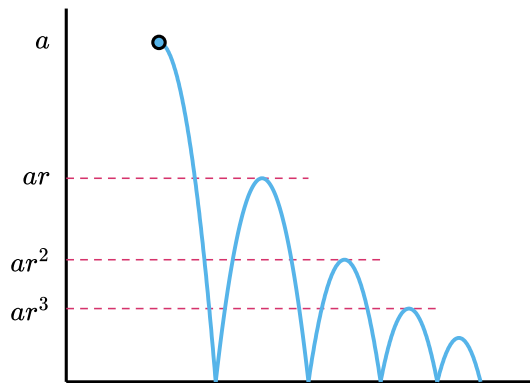


Figure 2.1.7 The ball falls a distance a , then rises and falls a distance ar , then ar^2 , and so on, so the total distance travelled is $a + 2ar + 2ar^2 + \dots$.

2.1.4 The n th Term Test

Theorem 2.1.8 If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Important note: If $\lim_{n \rightarrow \infty} a_n = 0$, we *cannot* conclude that $\sum_{n=1}^{\infty} a_n$ converges. See parts D and E in the example below. Also see [Example 2.1.12](#).

The following test is a consequence of the above theorem.

The n th Term Test.

If $\lim_{n \rightarrow \infty} a_n$ does not exist or $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Example 2.1.9 Testing series for convergence. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

A. $\sum_{n=1}^{\infty} n$

B. $\sum_{n=1}^{\infty} (-1)^n$

C. $\sum_{n=1}^{\infty} \frac{n^3 + 1}{n^2 + n}$

D. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

E. $\frac{4}{9} - \frac{8}{27} + \frac{16}{81} - \frac{32}{243} + \dots$

Solution.

A.

B.

C.

D.

E.



2.1.5 Combining Series

Theorem 2.1.10 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum ka_n = k \sum a_n = kA$ (Any number k).

Example 2.1.11 Using the difference rule. Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2^n}$.

Solution. (You have already summed both of these series in this section: one in [Subsection 2.1.2](#) and the other in part D of [Example 2.1.9](#).)

workspace



Example 2.1.12 A telescoping series with logarithms. Evaluate $\sum_{k=0}^{\infty} \ln\left(\frac{k+2}{k+1}\right)$, if it converges and if it diverges, show that it does.

Solution.

workspace



2.2 Taylor and Maclaurin Series

Objectives

After completing this section, you should be able to:

- Approximate a function near a point by its tangent line and by higher order polynomials built from its derivatives.
- Write down the Taylor series generated by f at $x = a$ and the Maclaurin series generated by f .
- Compute the Taylor polynomial of order n generated by f at $x = a$.
- Find Maclaurin series of standard functions such as $\sin(x)$ directly from the definition.

2.2.1 Definitions of Taylor Series, Maclaurin Series, and Taylor Polynomials

In this section we will answer the following question:

If all we know about a function is information at $x = a$, i.e. $f(a), f'(a), f''(a), \dots$, how can we approximate $f(x)$ with a polynomial $p(x)$?

These are skeletal notes: the definitions and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

Throughout, let f be a function with derivatives of all orders throughout some interval containing a as an interior point.

Before stating the definitions, let us see where the coefficients of such a polynomial have to come from. Suppose the only things we know about f are its readings at $x = a$: the value $f(a)$, the slope $f'(a)$, the second derivative $f''(a)$, and so on. We look for a polynomial written in terms of $x - a$,

$$p_n(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + \dots + c_n(x - a)^n, \quad (2.2.1)$$

and we pin down the unknown coefficients $c_0, c_1, \dots, c_n$ by requiring p_n to agree with f at $x = a$ in as many derivatives as it has coefficients. Writing it in terms of $x - a$ rather than x is what makes this manageable: every term after the first vanishes at $x = a$.

Work out the first few coefficients. In each row, differentiate the polynomial as many times as it has coefficients, evaluate at $x = a$, and match against f .

a line, $p_1(x) = c_0 + c_1(x - a)$: $c_0 =$ $c_1 =$

a parabola, $p_2(x) = c_0 + c_1(x - a) + c_2(x - a)^2$: $c_2 =$

a cubic, $p_3(x) = \dots + c_3(x - a)^3$: $c_3 =$

Notice what should *not* have happened: c_0 and c_1 should come out the same every time. Each new condition is the first one in which the next coefficient appears, so it determines that coefficient and leaves the earlier ones untouched. That is why p_2 is the tangent line with a single new term added to it, rather than a fresh approximation built from scratch. The pattern in the denominators should now be visible: $0!, 1!, 2!, 3!, \dots$

One computation settles every coefficient at once. Differentiate (2.2.1) exactly k times and evaluate at $x = a$. Say what happens to the terms of degree below k , to those of degree above k , and to the term $c_k(x - a)^k$ itself; then impose $p_n^{(k)}(a) = f^{(k)}(a)$ and solve for c_k .

$$p_n^{(k)}(a) =$$

$$c_k =$$

So there is nothing to choose. Once we ask a polynomial to match f and its first n derivatives at $x = a$, its coefficients are determined, and they are the numbers $f^{(k)}(a)/k!$. Reading $f^{(0)}$ as f and $0!$ as 1, the first coefficient $c_0 = f(a)$ fits the same formula. Letting n grow without bound leads to the following definitions.

Definition 2.2.1 Taylor Series. The **Taylor series generated by f at $x = a$** is

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots \\ &\quad + \frac{f^{(n)}(a)}{n!} (x-a)^n + \cdots \end{aligned}$$

◆

Definition 2.2.2 Taylor Polynomial of Order n . The **Taylor polynomial of order n generated by f at $x = a$** is the polynomial

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

◆

Remark 2.2.3 Maclaurin Series. The centre $a = 0$ is common enough to have its own name. The Taylor series generated by f at $x = 0$ is known as the **Maclaurin series generated by f** , which is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots.$$

Example 2.2.4 Approximating e^x near $x = 0$. Write down the Taylor polynomials of orders 1, 2, and 3 generated by $f(x) = e^x$ at $x = 0$.

Solution. Every derivative of e^x is e^x again, so the coefficients of Definition 2.2.2 are easy. Check your answers against Figure 2.2.5.

$$p_1(x) =$$

$$p_2(x) =$$

$$p_3(x) =$$

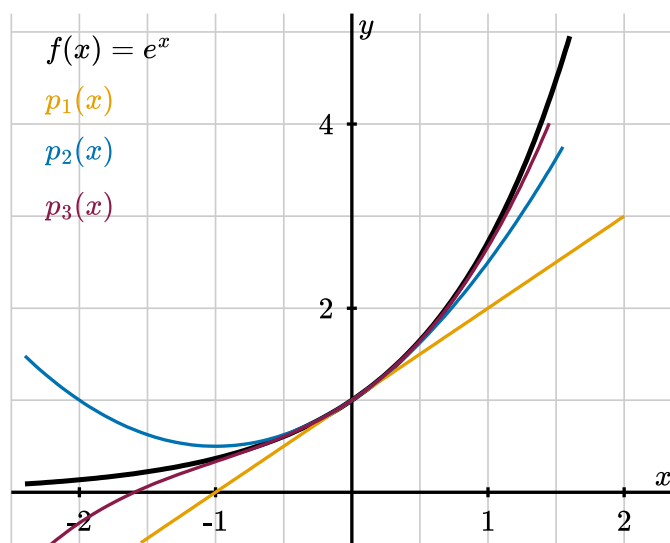


Figure 2.2.5 The function $f(x) = e^x$ together with the Taylor polynomials p_1 , p_2 , and p_3 at $x = 0$. Near $x = 0$ each polynomial hugs the curve more closely than the one before it.

An animation of this construction accompanies the online version of these notes, with sliders for the centre a and the order n . Two things are worth noting from it. Moving a rebuilds the same construction at a new point of the curve, where p_n again touches f . Raising n adds exactly one term, $\frac{f^{(n)}(a)}{n!}(x-a)^n$, which vanishes at $x = a$ — so it changes the shape of the polynomial everywhere else but never moves it off the point $(a, f(a))$.

The interval on which p_n stays within 0.1 of f widens with every term: about $[-0.5, 0.4]$ for p_1 , about $[-1.3, 1.2]$ for p_3 , and wider still by p_8 . So the Taylor polynomials do not merely improve at the centre; the region where they are usable grows. Whether that region eventually covers everything, and how large the error is at a given x , are the questions of the next section.

2.2.2 More Examples

Example 2.2.6 The Maclaurin series of $\sin(x)$. Consider the function $f(x) = \sin(x)$.

- A. Find the Taylor series generated by f at $x = 0$. Note that this is the same as the Maclaurin series generated by f .

- B. Calculate the first four Taylor polynomials $p_n(x)$.
- C. Plot the original function $f(x) = \sin(x)$ and the Taylor polynomials obtained in part B to confirm that the higher order polynomials provide a better approximation.
- A. Differentiate f repeatedly, evaluate at $x = 0$, find the pattern, then use [Remark 2.2.3](#).

derivatives:

at $x = 0$:

pattern:

series:

- B. Keep the first four non-zero polynomials.

$p_1(x) =$

$p_3(x) =$

$p_5(x) =$

$p_7(x) =$

- C. Check your polynomials against [Figure 2.2.7](#).



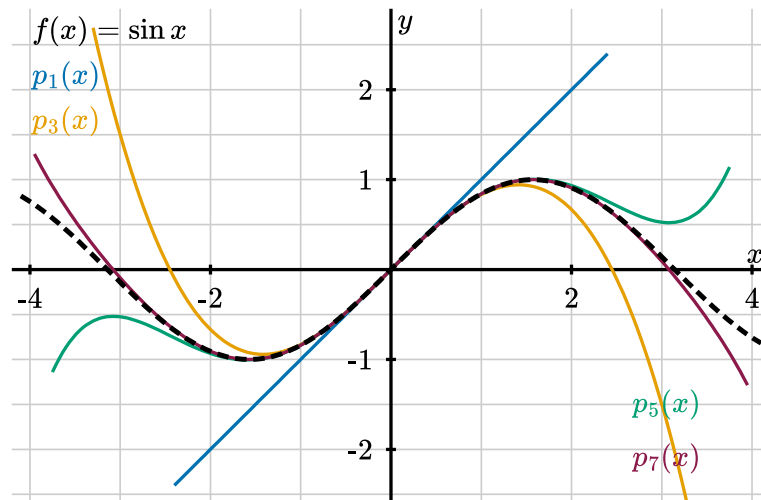


Figure 2.2.7 The function $f(x) = \sin x$ together with the Taylor polynomials at $x = 0$, where $p_1(x) = x$, $p_3(x) = x - \frac{x^3}{3!}$, $p_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$, and $p_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$.

Checkpoint 2.2.8 Consider the function $f(x) = \frac{x}{1 - 9x}$.

- Compute the Maclaurin series generated by f . Express the result in sigma notation.
- Find the interval of convergence for this series.

A.

B.

Hint. Write $\frac{1}{1 - 9x}$ as a geometric series first.

Checkpoint 2.2.9 Compute the Taylor series of $f(x) = \frac{1}{3x + 7}$ at $x = 1$.

workspace

Hint. Either differentiate repeatedly at $x = 1$, or write $3x + 7$ in terms of $x - 1$ and reduce to a geometric series.

2.3 Taylor Series and its convergence

Objectives

The main objectives of this section are to answer the following questions:

1. Under what conditions is the function $f(x)$ equal to the Taylor series generated by f at $x = a$? That is, when can we say $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$?
2. If the Taylor polynomial of degree n , $p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$, is used as an approximation to $f(x)$, how can we estimate the error in this approximation? That is, how can we estimate the magnitude of $R_n(x) = f(x) - p_n(x)$?

[Theorem 2.3.1](#) and [Theorem 2.3.4](#) will enable us to answer these questions. We begin by stating Taylor's formula and the remainder estimation theorem, and then use them to bound the error in a Taylor approximation, discuss some applications of these theorems, and close with the proof of the remainder theorem.

These are skeletal notes: the theorems and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

2.3.1 Taylor's Formula and The Remainder Estimation Theorem

Theorem 2.3.1 Taylor's Formula. *Let f be a function that has $n + 1$ continuous derivatives on an open interval I containing a . Then for each $x \in I$ and for each positive integer n , there exists a number c between a and x such*

that

$$f(x) = p_n(x) + R_n(x), \quad (2.3.1)$$

where

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k, \quad (2.3.2)$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}. \quad (2.3.3)$$

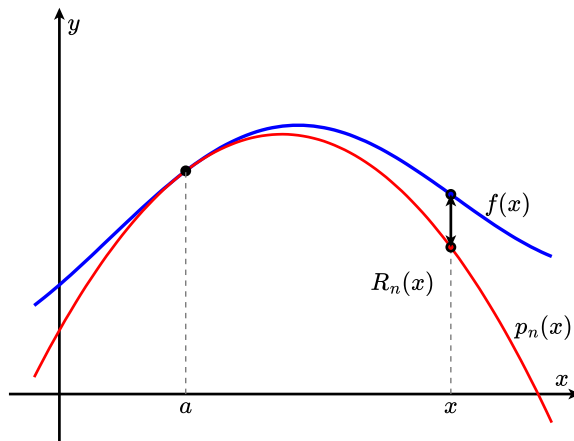


Figure 2.3.2 Taylor's Formula: $f(x) = p_n(x) + R_n(x)$. The Taylor polynomial p_n agrees with f at $x = a$; away from a , the vertical gap between them is the remainder $R_n(x)$.

Remark 2.3.3 Finding an upper bound for the error term without knowing the value of c . Usually the value of c is not explicitly known. However, we may manage to find **an upper bound** for the error term $R_n(x)$ without knowing the exact value of c . This is achieved by finding an upper bound M for $|f^{(n+1)}(c)|$, where c is between a and x , and then using this upper bound to estimate the error term.

This is the idea behind every error estimate in this section. It is used in [Subsection 2.3.2](#) to prove that the Taylor series of e^x converges to e^x for every x , and in each of the three examples of [Subsection 2.3.3](#).

Theorem 2.3.4 The Remainder Estimation Theorem. *Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a . Then for each $x \in I$ and for each positive integer n , there exists a number c between a and x such that*

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}, \quad (2.3.4)$$

where M is an upper bound for $|f^{(n+1)}(c)|$ on the interval between a and x .

(Derive (2.3.4) from (2.3.3).)

workspace

You are not expected to know the proof of Taylor's formula itself, only to understand the statement and its implications. The proof is worked through in [Subsection 2.3.5](#).

2.3.2 An example of a Taylor series that converges

Consider the function $f(x) = e^x$. The Taylor series generated by f at $x = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \text{_____}, \quad (2.3.5)$$

and its remainder is

$$R_n(x) = \text{_____}. \quad (2.3.6)$$

(Compute [\(2.3.5\)](#) and [\(2.3.6\)](#).)

Taylor series of e^x at $x = 0$:

$R_n(x) =$

(Show that [\(2.3.5\)](#) converges to e^x for every $x \in \mathbb{R}$.)

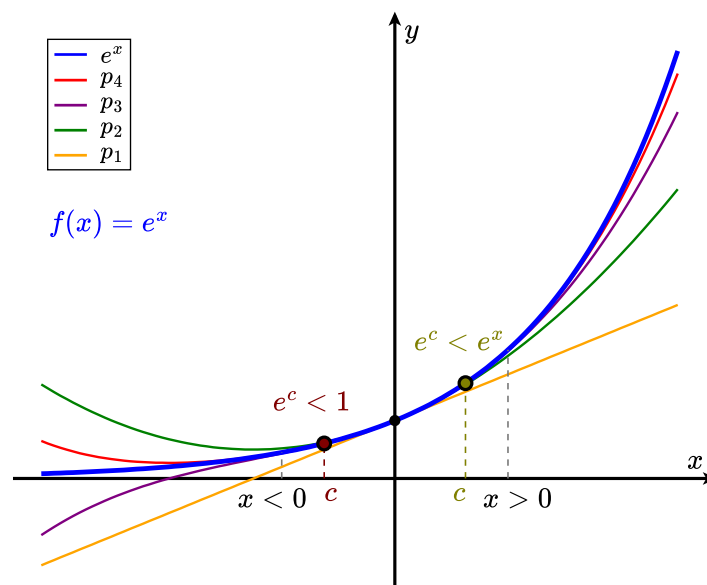
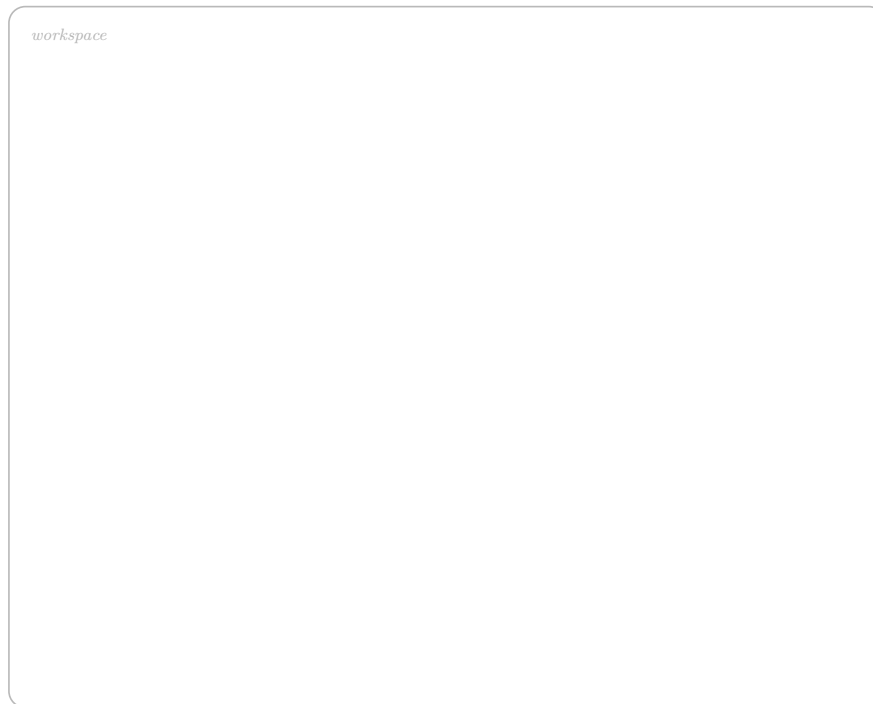


Figure 2.3.5 Schematic graph of $f(x) = e^x$ and its Taylor polynomials $p_1, \dots, p_4$ about 0.

An animation of this convergence — the Taylor polynomials p_n hugging the graph of $f(x) = e^x$ over a wider and wider interval as n grows — accompanies the online version of these notes.

2.3.3 The error in using a Taylor polynomial

Suppose we approximate $f(x)$ by the Taylor polynomial (2.3.2) of degree n at $x = a$. The error is $R_n(x) = f(x) - p_n(x)$, which is (2.3.1) rearranged, and is given by (2.3.3). To bound it we need an upper bound M for $|f^{(n+1)}(c)|$ on the interval between a and x , as in Remark 2.3.3. The three examples below

carry out that step in three different settings.

Example 2.3.6 Approximating e^x using a Taylor polynomial. Approximate $e^{0.5}$ using the Taylor polynomial of degree 2 at $x = 0$. Find an upper bound for the error in this approximation.

Solution.

$p_2(x) =$

bound on $|R_2(0.5)|:$

actual error:



Example 2.3.7 Given a desired error bound, find values of x for which the approximation is guaranteed to be valid. Assume that we use the Taylor polynomial of degree 3 at $x = 0$ to approximate $\sin(x)$. For approximately what values of x can you replace $\sin(x)$ by such a Taylor polynomial with an error of magnitude no greater than 4×10^{-3} ?

Solution.

$$p_3(x) = \quad \text{and} \quad R_4(x) =$$

bound on $|R_4(x)|$:

solve for $|x|$:



Example 2.3.8 Finding the value of n such that the error is less than a given tolerance. Find the smallest value of n for which the polynomial approximation for $\sin(x)$ is accurate to ± 0.005 for values of x in the interval $[-\pi/2, \pi/2]$.

Solution.

bound on $|R_n(x)|$:

inequality to solve:

test values of n :



2.3.4 Applications of Taylor Series and Remainder Theorem

Physicists often use Taylor series to approximate functions in order to simplify calculations. The first example is the kinetic energy of an object in relativistic mechanics. The second example is the approximation of the period of a pendulum. We will discuss the error in using these approximations and when it is valid to use them. Additionally, we will discuss how to use the Taylor series to estimate the value of π in the last example.

2.3.4.1 Approximating Relativistic Kinetic Energy

In relativistic mechanics, the mass of an object moving with velocity v is given by:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (2.3.7)$$

where m_0 is the rest mass of the object and c is the speed of light. Then the kinetic energy of an object of mass m moving with velocity v is:

$$K = mc^2 - m_0c^2 \quad (2.3.8)$$

In the case when $v \ll c$, we can use the Taylor series to approximate the kinetic energy. In [Example 2.3.10](#) we will show that the kinetic energy can be approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$. See [Figure 2.3.9](#) for a comparison of the relativistic kinetic energy and its Newtonian approximation.

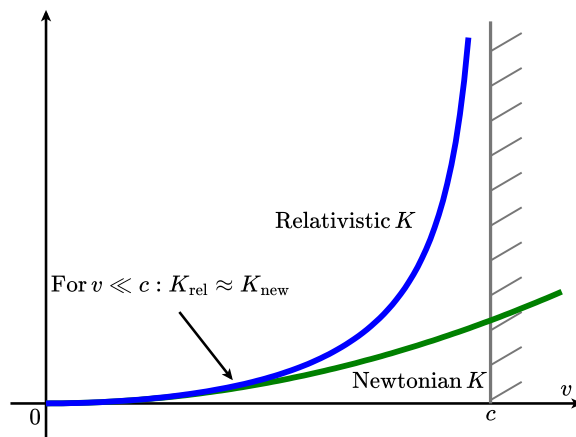


Figure 2.3.9 Relativistic versus Newtonian kinetic energy. The relativistic energy races toward a wall at the speed of light c , while the Newtonian energy follows a gentle parabola; the two are approximately in agreement only when $v \ll c$.

Example 2.3.10 Newtonian kinetic energy as an approximation to relativistic kinetic energy when $v \ll c$. Show that the kinetic energy of an object moving with velocity v can be approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$.

Solution.

workspace

When $v \ll c$, we can therefore approximate the kinetic energy as:

$$K \approx \frac{1}{2}m_0v^2. \quad (2.3.9)$$



Example 2.3.11 Estimating the error in using the Newtonian kinetic energy formula to approximate the relativistic kinetic energy. Assume that a car is moving with a velocity of $v = 30m/s$ (≈ 67 miles per hour). Use the remainder's theorem to estimate the error in using the Newtonian kinetic energy formula (2.3.9) to approximate the relativistic kinetic energy of the car. The speed of light is $c = 3 \times 10^8m/s$.

Solution.

workspace



2.3.4.2 Small-Angle Approximation for a Pendulum

We begin this section by briefly reviewing the forces acting on a simple pendulum and how the small-angle approximation allows us to treat its motion as simple harmonic motion.

The bob moves along the arc, so only the component of gravity *tangent* to that arc drives the motion. Resolving the weight $m\vec{g}$ into a component along the string ($mg\cos\theta$, balanced by the string tension $\vec{F}_T$) and a component tangent to the arc, as shown in [Figure 2.3.12](#), gives the restoring force

$$F = -mg\sin\theta. \quad (2.3.10)$$

The minus sign indicates that the force always points back toward the equilibrium (straight-down) position. This is *not* Hooke's law: the force is proportional to $\sin\theta$, not to the displacement itself, so the motion is not exactly simple harmonic. Writing the arc displacement as $x = \ell\theta$, we would need F to be proportional to x —that is, to θ —for the motion to be simple harmonic.

The small-angle approximation bridges this gap. From the Taylor series

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots, \quad (2.3.11)$$

when θ is small (in radians) the higher-order terms are negligible and $\sin\theta \approx \theta$. The restoring force then becomes

$$F \approx -mg\theta = -\frac{mg}{\ell}x,$$

which *is* Hooke's law $F = -kx$ with effective spring constant $k = \frac{mg}{\ell}$.

This is exactly what fixes the period. Newton's second law $F = ma = m\frac{d^2x}{dt^2}$ turns Hooke's law into the equation of motion

$$m\frac{d^2x}{dt^2} = -\frac{mg}{\ell}x, \quad \text{that is,} \quad \frac{d^2x}{dt^2} = -\frac{g}{\ell}x.$$

Notice that the mass cancels. The equation says that x is a function whose second derivative is a negative multiple of itself, and the functions with that property are the sines and cosines: writing $\omega^2 = \frac{g}{\ell}$, every solution has the form

$$x(t) = A\cos(\omega t) + B\sin(\omega t), \quad \omega = \sqrt{\frac{g}{\ell}},$$

which you can verify by differentiating twice. The number ω is the angular frequency, and $\cos(\omega t)$ and $\sin(\omega t)$ repeat when ωt increases by 2π . So the motion repeats after a time T with $\omega T = 2\pi$, giving

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{\ell}{g}}. \quad (2.3.12)$$

Thus, for small swings the pendulum behaves as a simple harmonic oscillator, with the period [\(2.3.12\)](#), which is independent of both the amplitude and the mass. Taylor's Remainder Theorem (see [Example 2.3.13](#)) enables us to quantify how small θ must be for this approximation.

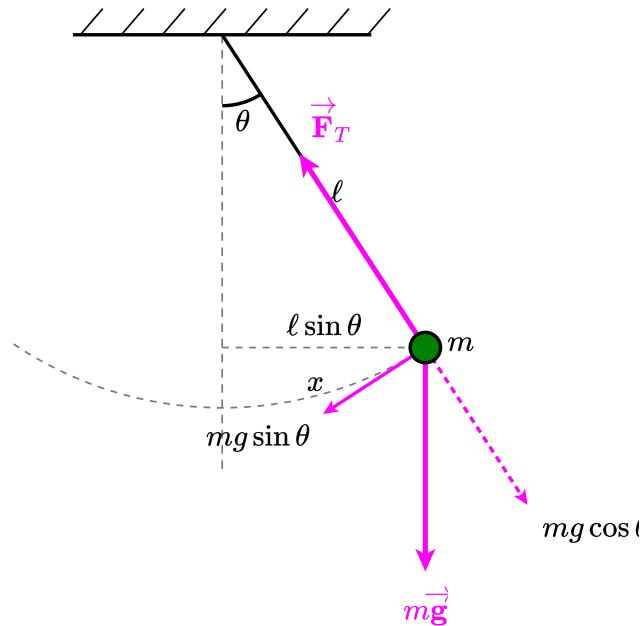


Figure 2.3.12 Forces on a simple pendulum. The weight $m\vec{g}$ resolves into a component $mg \cos \theta$ along the string (balanced by the tension $\vec{F}_T$) and a component $mg \sin \theta$ tangent to the arc, which acts as the restoring force.

An animation of these forces — the weight resolving into its radial and tangential components as the pendulum swings — accompanies the online version of these notes.

Example 2.3.13 Simple Harmonic Motion of a Pendulum as an Approximation. Use the remainder theorem to analyze the claim made in Giancoli’s textbook that for small angles, $\sin \theta \approx \theta$. Specifically, show that the error is less than 1% for angles below 15° .

Here is the exact quote from Giancoli’s textbook:

“For angles less than 15° , the difference between θ (in radians) and $\sin \theta$ is less than 1%.”

Solution.

workspace



2.3.4.3 Approximating the value of π using Taylor series

In this subsection, we will discuss how to approximate the value of π using Taylor series. We will use the Taylor series for $\arctan(x)$, which centered at $a = 0$ is

$$\begin{aligned}\arctan(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \quad |x| \leq 1.\end{aligned}\tag{2.3.13}$$

Example 2.3.14 Approximating π using the Taylor series for $\arctan(x)$. Use the Taylor series for $\arctan(x)$ to approximate the value of π . Use Taylor's Remainder Theorem to find an upper bound for the error in this approximation, and show that the error decreases as the order n of the Taylor polynomial increases.

Solution.

workspace

For the record: the derivatives of $f(x) = \arctan(x)$ satisfy

$$\left| f^{(n+1)}(t) \right| \leq n! \quad \text{for all } t \in \mathbb{R}, \quad (2.3.14)$$

and the resulting error bound is

$$|\pi - 4T_n(1)| \leq \frac{4}{n+1}. \quad (2.3.15)$$

The bound goes to zero as $n \rightarrow \infty$, but very slowly: to guarantee an error of at most 0.01 one needs $n \geq 399$. $\blacktriangle$

As we saw in the previous example, Taylor series approached π very slowly. In the project below, we will see how to use the so-called Euler's formula to approximate π much faster. As you may know, there are many other methods to approximate π , which we will not cover here.

Project 2.3.4.1 Computing π with Euler's identity. In [Example 2.3.14](#) we approximated π by evaluating the Taylor series (2.3.13) at $x = 1$, and Taylor's Remainder Theorem gave the error bound (2.3.15), which decreases very slowly. In this guided problem we compute π far more efficiently using **Euler's identity**

$$\arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}, \quad (2.3.16)$$

which lets us evaluate the Taylor series at the small arguments $x = \frac{1}{2}$ and $x = \frac{1}{3}$, where it converges much faster.

(a) *Proving Euler's identity.*

Let $\alpha = \arctan\left(\frac{1}{2}\right)$ and $\beta = \arctan\left(\frac{1}{3}\right)$. Use the addition formula

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

to prove that $\alpha + \beta = \frac{\pi}{4}$.

$\tan(\alpha + \beta) =$

why $\alpha + \beta$ must be $\pi/4$ and not another angle:

(b) *The approximation.*

Let $T_n(x)$ be the Taylor polynomial of order n for $\arctan(x)$ centered at 0. Use Euler's identity (2.3.16) to explain why

$$\pi \approx 4 \left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right) \right].$$

Write out this approximation explicitly for $n = 3$.

why the approximation holds:

the case $n = 3$, as a number:

(c) *Bounding the error with the Remainder Theorem.*

In Example 2.3.14 we showed that the derivatives of $f(x) = \arctan(x)$ satisfy (2.3.14). Use (2.3.4) to show that for $|x| \leq 1$

$$|\arctan(x) - T_n(x)| \leq \frac{|x|^{n+1}}{n+1},$$

and conclude that

$$|\pi - 4 \left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right) \right]| \leq \frac{4}{n+1} \left[\left(\frac{1}{2}\right)^{n+1} + \left(\frac{1}{3}\right)^{n+1} \right].$$

the bound on $|\arctan(x) - T_n(x)|$:

combining the two remainders:

(d) How much better is it?

Evaluate the error bound for $n = 9$ and compare it with the bound $\frac{4}{10} = 0.4$ obtained in [Example 2.3.14](#) for the same order. Then find the smallest n for which the bound guarantees an error of at most 10^{-6} .

the bound at $n = 9$, versus 0.4:

the smallest n with bound at most 10^{-6} :

An animation of the approximation $4 \left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right) \right]$ converging to π , with its error compared against the much slower method at $x = 1$, accompanies the online version of these notes.

2.3.5 Proof of the remainder theorem

In this subsection, we will provide a proof of the remainder theorem.

Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a . We want to show that for each $x \in I$ and for each positive integer n , there exists a number c between a and x such that

$$f(x) = p_n(x) + R_n(x),$$

where

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k,$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

To prove this, we will first prove the following lemma:

Lemma 2.3.15 *Let $f(x)$ be a function that is $(n+1)$ -times differentiable. Also, suppose that $f(a) = f'(a) = f''(a) = \dots = f^{(n)}(a) = 0$ and $f(x) \neq 0$, where*

$x \neq a$. Then there exists a number c between a and x such that $f^{(n+1)}(c) = 0$.

Proof.

workspace

Remark 2.3.16 The error function and its derivatives are zero at the point of expansion. Note that for the error function $R_n(x) = f(x) - p_n(x)$, we have $R_n(a) = R'_n(a) = R''_n(a) = \dots = R_n^{(n)}(a) = 0$.

Completing the proof.

workspace

Chapter 3

Assignments and Review Problems

The ten assignments and the four review problem sets, in one place. Worked solutions appear beneath each problem once the assignment's due date has passed; solutions for the review sets are posted the same way.

Using the AI Tutor.

Stuck? A Socratic AI tutor is available on every assignment page. It knows the problems on the page and will coach you with questions and gentle hints — but it will never give away an answer. Here is how to use it.

Opening the tutor. On an assignment page, click the round “Tutor” button in the lower-right corner. A chat panel opens and stays in place as you scroll, so you can read a problem and talk to the tutor at the same time. The dash button in the panel's title bar minimizes it back to the button.

The tutor is available on each of the ten assignment pages. The review problem sets do not have it: those sets are practice for the exams, where you will be entirely on your own.

One-time setup. The first time you open the tutor it asks for a free Google AI Studio key (this takes about a minute):

1. Go to aistudio.google.com/apikey and sign in with a *personal* Google account — a regular Gmail. Do *not* use your @scu.edu account: the university has this service turned off, and you will see “Your account is managed by an organization.”
2. Click *Create API key* and copy it.
3. Paste it into the tutor.

The key is stored only in your own browser and is sent only to Google; this site never sees it. Treat the key as yours alone: anyone who has it can spend your free quota, so do not share it or paste it anywhere else.

Privacy: what to keep out of the chat. Everything you type into the tutor is sent to Google and handled under Google's terms, not the university's. On the free tier your conversations *may be read by people at Google* and used to improve their products, and they are not

covered by the protections your school records have. Nothing you type is private, so treat the panel as a public place and keep it to mathematics.

- Do not type anything that identifies you: your name, student ID, SCU email, phone number, address, or date of birth. The tutor does not need to know who you are in order to help with a gradient.
- Do not type anything about anyone else — a classmate’s name, their work, or their grade. That is their information to share, not yours.
- Keep out anything sensitive or unrelated to the course: health, finances, immigration or visa status, family matters, passwords, or anything you would not want quoted back to you.
- Before you paste or upload, look at what you are sending. A screenshot or a photo of your work can carry your name, a page header, or a browser tab in the background.
- The tutor is a study aid, not a counselor and not the instructor. If something personal is affecting your work in this course, bring it to office hours or to the university’s student services instead — those conversations belong with a person, and they stay confidential.

None of this is a reason to avoid the tutor. Ask it anything about the mathematics — what you tried, where you are stuck, why a step feels wrong. Just keep yourself, and everyone else, out of the transcript.

Working with the tutor. Choose the problem you are working on from the drop-down at the top of the panel, then tell the tutor what you have tried so far — the more of your own thinking you show, the more useful its questions become. The tutor can nudge you toward a starting point, and it can point you to sections and worked examples in these notes to study.

If your answer is wrong, the tutor will tell you so and help you understand *why* it is wrong: which step breaks, what misconception is behind it, and what picture or small example will let you see the failure for yourself. Note that this requires showing your reasoning — a bare answer with no work, right or wrong, will simply get asked for the thinking behind it. And what the tutor will not do is tell you how to correct an error — no fixed-up step, no “instead, you should...” Finding the right path once you understand why the old one fails is your part of the work.

Two models. Next to the problem picker is a “Model” drop-down offering *Careful* and *Quick*. Both are Gemini Flash models. Careful is the default: it thinks before it answers, so it takes a few seconds longer and is better at staying in the role of a tutor rather than a solutions manual. Quick is lighter and replies almost instantly. Either will coach you and neither will hand you an answer; switching between them costs nothing and does not disturb your conversation or your hint count. Try both — and tell me which you prefer.

Hints are limited on purpose. The point of the tutor is to get you unstuck, not to walk you to the answer, so hints are deliberately scarce. The exact rules:

- You get at most *three hints per problem, total* — the budget never resets. The panel always shows how many you have left on the problem you selected (“Hints left: 2/3”).
- *Every* reply from the tutor costs a hint — there are no free ones. Lead with what you have tried, so each reply can meet your actual thinking instead of spending itself asking for it.
- Restarting the conversation, reloading the page, or switching problems does not refill hints. Spend them thoughtfully — full solutions are posted after the due date.
- When your hints for a problem run out, bring your work to office hours — working through it together is exactly what office hours are for.

Tell me how it is going. Under every hint is a single question — “Was this hint helpful?” — with a thumb up and a thumb down. One tap is the whole thing, and it is the most useful second you can spend on this course: it tells me which hints land and which ones miss, hint by hint. Nothing but your answer and which problem you were on is sent.

For anything longer there is the “feedback” button at the top of the panel, which sends me a short note about how the tutor is working for you: whether it helped, and — in a box you can write as much as you like in — what would make it better. Tapping the thumb down offers it too. All of it is anonymous, it has nothing to do with your grade, and the tutor itself never sees it. If you want me to see the exchange you just had, there is a box to attach it; it is off unless you tick it. The tutor is new, and what you write is how it gets better.

Finally, know what the tutor will and will not do. It will not state answers, complete a solution, or turn a wrong answer into a right one — no matter how you ask. There is exactly one way to get an answer confirmed: state your conclusion *together with your own justification*. If your reasoning is sound and complete, the tutor will tell you that you have it; if there is a gap, it will point at the gap and let you close it. A bare answer, right or wrong, earns nothing.

3.1 Assignment 1

These problems exercise the hyperbolic identities, the derivatives of the hyperbolic functions and the inverse hyperbolic functions. If you would like to review the material first, see [Hyperbolic Functions](#).

1. Show that $\tanh^2(x) + \operatorname{sech}^2(x) = 1$, for all real numbers x .

Solution. We write both terms over the common denominator $\cosh^2(x)$ and use the identity $\cosh^2(x) - \sinh^2(x) = 1$, i.e. $\sinh^2(x) + 1 = \cosh^2(x)$:

$$\begin{aligned}\tanh^2(x) + \operatorname{sech}^2(x) &= \frac{\sinh^2(x)}{\cosh^2(x)} + \frac{1}{\cosh^2(x)} \\ &= \frac{\sinh^2(x) + 1}{\cosh^2(x)} = \frac{\cosh^2(x)}{\cosh^2(x)} = 1.\end{aligned}$$

Note that this is valid for every real x , since $\cosh(x) \geq 1 > 0$ and so the denominator is never zero.

2. Compute $\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right)$.

Solution. We use $\frac{d}{dx} \sinh(u) = \cosh(u) \frac{du}{dx}$ with $u = \frac{3}{x} = 3x^{-1}$, so $\frac{du}{dx} = -3x^{-2} = -\frac{3}{x^2}$. Hence

$$\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right) = 3 \cosh\left(\frac{3}{x}\right) \left(-\frac{3}{x^2}\right) = -\frac{9}{x^2} \cosh\left(\frac{3}{x}\right).$$

3. Simplify $\cosh(\ln 2)$.

Solution. Using the definition $\cosh x = \frac{e^x + e^{-x}}{2}$ together with $e^{\ln 2} = 2$ and $e^{-\ln 2} = \frac{1}{e^{\ln 2}} = \frac{1}{2}$,

$$\cosh(\ln 2) = \frac{e^{\ln 2} + e^{-\ln 2}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{\frac{5}{2}}{2} = \frac{5}{4}.$$

4. Solve the equation $5 \sinh(x) - 3 \cosh(x) = 3$ for x .

Solution. We first replace the hyperbolic functions by their definitions:

$$\begin{aligned}5 \sinh(x) - 3 \cosh(x) &= 3 \\ \Rightarrow \frac{5(e^x - e^{-x})}{2} - \frac{3(e^x + e^{-x})}{2} &= 3 \\ \Rightarrow e^x - 4e^{-x} &= 3.\end{aligned}$$

Multiplying through by e^x and writing $u = e^x$ turns this into a quadratic equation:

$$\begin{aligned}u^2 - 3u - 4 &= 0 \\ \Rightarrow (u - 4)(u + 1) &= 0.\end{aligned}$$

Since $u = e^x$ is positive, the root $u = -1$ is impossible, and only $u = 4$ survives. Therefore

$$e^x = 4 \quad \Rightarrow \quad x = \ln 4 = 2 \ln 2.$$

5. Following the method of [Example 1.4.2](#), show that the inverse hyperbolic tangent is given by

$$\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad -1 < x < 1.$$

Solution. Unlike $\cosh$, the function $\tanh$ is increasing on all of $\mathbb{R}$, so no restriction of its domain is needed. Set $y = \tanh(x)$ and solve for x . Multiplying the numerator and the denominator by e^x gives

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot \frac{e^x}{e^x} = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

Writing $u = e^{2x}$ and clearing the denominator,

$$y(u + 1) = u - 1$$

$$\begin{aligned}\Rightarrow y + 1 &= u - yu = u(1 - y) \\ \Rightarrow u &= \frac{1 + y}{1 - y}.\end{aligned}$$

Therefore $e^{2x} = \frac{1 + y}{1 - y}$, and taking the natural logarithm gives $2x = \ln\left(\frac{1 + y}{1 - y}\right)$. Interchanging the names of the two variables, so that $y = \tanh^{-1}(x)$ is the inverse function,

$$y = \tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right).$$

The range of $\tanh$ is the interval $(-1, 1)$, which is exactly the set of x for which $\frac{1+x}{1-x}$ is positive, so this is the domain of $\tanh^{-1}$.

6. Use the substitution $x = \tanh(t)$ to calculate

$$\int \frac{dx}{1 - x^2}, \quad -1 < x < 1,$$

and then use [Worksheet Exercise 3.1.5](#) to write your answer with a logarithm. *Hint:* $\frac{d}{dt} \tanh(t) = \operatorname{sech}^2(t)$, and $1 - \tanh^2(t) = \operatorname{sech}^2(t)$.

Solution. With $x = \tanh(t)$ we have $dx = \operatorname{sech}^2(t) dt$, and the denominator collapses by the identity in the hint:

$$\begin{aligned}\int \frac{dx}{1 - x^2} &= \int \frac{\operatorname{sech}^2(t)}{1 - \tanh^2(t)} dt \\ &= \int \frac{\operatorname{sech}^2(t)}{\operatorname{sech}^2(t)} dt \\ &= \int dt = t + C.\end{aligned}$$

Since $x = \tanh(t)$ means $t = \tanh^{-1}(x)$, and using [Worksheet Exercise 3.1.5](#) to rewrite the inverse function,

$$\begin{aligned}\int \frac{dx}{1 - x^2} &= \tanh^{-1}(x) + C \\ &= \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right) + C.\end{aligned}$$

This agrees with the answer obtained by partial fractions, since $\frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2} \ln(1 + x) - \frac{1}{2} \ln(1 - x)$.

3.2 Assignment 2

These problems work with sequences, finite sums and geometric series, ask whether a series converges and what it sums to, and then use the Maclaurin series of the standard functions to recognise a sum. If you would like to review the material first, see [Infinite Series](#) and [Taylor and Maclaurin Series](#).

1. The terms of a geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ satisfy

$$a_5 + a_7 = 1500 \quad \text{and} \quad a_{11} + a_{13} = 187500,$$

where $a_n = ar^{n-1}$ denotes the n th term. Find $a_1 + a_2 + a_3$.

2. What is the sum of all multiples of 7 or 11 less than 1000?
 3. What can we conclude by applying the n th term test in the following series?

A. $\sum_{n=1}^{\infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n}$

B. $\sum_{n=1}^{\infty} \frac{2\ln(n)}{3n}$

4. Evaluate the following sums or show that they diverge.

A. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

B. $\sum_{n=0}^{\infty} e^{-2n}$

C. $\sum_{n=0}^{\infty} e^{2n}$

D. $\sum_{n=1}^{\infty} \left(1 + \frac{4}{n}\right)^n$

E. $\sum_{n=2}^{\infty} \frac{2}{3^n}$

5. Answer questions A and B below for the following infinite series:

$$\sum_{k=1}^{\infty} \left(\frac{6}{k^2} - \frac{6}{(k+1)^2} \right).$$

A. Does the n th-term test apply? Remember to fully justify your answer.

B. Evaluate the series or show that it diverges.

6. Express 8.81818181818... as a rational number, i.e. in the form $\frac{p}{q}$, where p and q are positive integers with no common factors.

7. Find the sum of the convergent series $\sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}$.

8. Find the sum of the series $\sum_{n=0}^{\infty} \frac{1}{n!}$.

9. Find the sum of the series

$$5 + 5^2 + \frac{5^3}{2!} + \frac{5^4}{3!} + \cdots + \frac{5^{n+1}}{n!} + \cdots$$